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The palaeomagnetic record of the Santiago volcanics (Republic of Cape Verde); multiphase magnetization and age consideration

M.C. Abranches [1], K.M. Storetvedt [2]*, A. Serralheiro [3] and R. Løvlie [4]

[1] *Departamento de Fisica, Universidade de Lisboa, P-1200 Lisboa (Portugal)*
[2] *Department of Physics, University of Newcastle upon Tyne, Newcastle upon Tyne NE1 7RU (U.K.)*
[3] *Departamento de Geologia — Centro de Vulcanologia, Faculdade de Ciencias, Universidade de Lisboa, P-1200 Lisboa (Portugal)*
[4] *Geofysisk Institutt, Universitetet i Bergen, N-5007 Bergen (Norway)*

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ABSTRACT

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The palaeomagnetism of the older volcanic complex of Santiago constitutes a multicomponent structure, and magneto-mineralogical evidence suggests that the various remanence vectors are carried principally by titanomaghaemites. When viewed in the context of the African palaeomagnetic reference frame the predominant magnetization is consistent with an Upper Tertiary age, but two older magnetizations of inferred Lower Tertiary age are also in evidence. The demonstrable secondary nature of all three palaeomagnetic axes (each of them represented by both polarities) makes it likely that the actual emplacement of the older complex is of pre-Tertiary age, being probably an integral component of the pronounced tectonomagmatic uplift phase that affected the entire northwest African margin in Upper Cretaceous time. The older complex of Santiago can probably be correlated with the igneous core of the neighbouring island of Maio, which apparently formed during Turonian time (~ 90 Ma ago). Some younger lavas of Upper Miocene age (the Flamengos Formation) show a much simpler palaeomagnetic history than rocks of the older complex, yielding only the Upper Tertiary palaeomagnetic field axis.

1. Study perspectives

The Cape Verde Archipelago sits on an extensive oceanic platform, elevated some 5.5 km above the abyssal plain, between the continental rise of Senegal/Mauritania and the Mid-Atlantic Ridge. This major topographic structure, presumably of volcanic origin, is of paramount importance in any theory on the development of the Central Atlantic crust. Based on geological and palaeomagnetic studies on the island of Maio (Serralheiro, 1970; Stillman et al., 1982; Storetvedt and Løvlie, 1983) it appears well established that the early history of this island goes back to the Upper Cretaceous, during which major tectonomagmatic forces uplifted the local oceanic crust, exposing deep-sea fossiliferous Jurassic and Lower Cretaceous beds to subaerial conditions. In fact, the early geological history of Maio and Fuerteventura (Canary Islands) is very similar and appears to mirror the Cretaceous development of the entire northwest African margin (Storetvedt, 1987).

The present palaeomagnetic investigation of Santiago is a further contribution to the question of the early geological history of the Cape Verde Islands. Santiago is dominantly formed by volcanics, which can be subdivided into a series of younger strata (Miocene–Quaternary in age) and an inner older volcanic complex. The island constitutes a large ancient volcano with long-lasting

* On leave from: Geofysisk Institutt, Universitetet i Bergen, N-5007 Bergen, Norway.

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but intermittent activity. It is not surprising, therefore, that the K–Ar dating technique has been successfully applied only to the younger volcanic strata, substantial argon loss and radiometric age resetting being inferred for the older complex (Bernard-Griffiths et al., 1975). Despite the problems of thermo-chemical palaeomagnetic overprinting of the older rocks, potentially giving rise to extensive magnetic resetting, thorough examination of remanence data may give a further clue to the age problem of the island's early volcanic history. It is this age aspect that forms the primary goal of the present study.

2. Outline geological history; palaeomagnetic sampling

The major knowledge of the stratigraphy and petrology of the island is based on work by Serralheiro (1970, 1976), Silva and Figueiredo (1976) and Silva (1977). The oldest exposed rock association, the inner eruptive complex, consists of a variety of intricately interwoven extrusive and intrusive rocks, for which the present outcrop pattern is delineated in Fig. 1. The oldest exposed rocks are (1) an association of submarine and subaerial basaltic lava flows and an accompanying volcanic plug system, ranging in composition from ankaratrites to basanites (Alves et al., 1979). Then follows (2) small irregular bodies and pipes of feldspathoidal gabbros, pyroxenites, syenites, etc., (3) intravolcanic breccia and breccia type dykes, (4) a series of phonolitic and trachytic intrusions and extrusions, and finally (5) carbonatite intrusions (Alves et al., 1979). Carbonatites are also common in association with the alkaline coarser-grained rocks (2) and with the intravolcanic breccia (3). The relationship and distinction between the different 'units' of the older complex is not always clear.

After an erosional–tectonic unconformity (including local conglomerate formation), the older extrusive–intrusive complex was followed by an extensive sequence of pillow lavas, the Flamengos Formation, now exposed only along the periphery of the island. Next in the stratigraphic column comes a sequence of sediments, terrigenous and marine conglomerates, and fossiliferous limestones and calcarenites. Then follows the very important Pico da Antónia Complex consisting of a thick lava pile that covers the greater part of the island. Material from this complex forms all of the higher topography (maximum elevation of 1394 m), and at the peak of its activity, in Lower–Middle Pliocene time, the Pico da Antónia eruptive phase must have turned the whole island into a quite impressive volcano. The terminal phases of volcanism in Santiago are represented by the Assomada Formation, filling topographic lows in the interior of the island, and by the Monte das Vacas Formation constituting some 50 minor and widely scattered cones of scoria and pyroclasts. The main petrographic column of the island is shown in Fig. 2.

The volcanism in Santiago is of the fissural type and, in the older complex in particular, feeder dykes and plugs form a very dense pattern. The density of these intrusions is sometimes so high that the pre-intrusive rock structures are completely obliterated. In certain cases it may also be difficult to distinguish between older and younger intrusives. It appears obvious, therefore, that the older volcanic complex must have suffered a lengthy and extensive thermo-chemical overprinting. There has also been fumarolic activity so that these rocks have undergone a different type and higher degree of secondary alteration than the younger series. The more advanced state of oxidation generally recognized in the older complex may indeed have formed the necessary chemical barrier against complete remagnetization during subsequent magmatic events (such as the very extensive Pico da Antónia volcanism).

The rock material for the palaeomagnetic study was collected at 51 sites encompassing altogether 185 samples. Forty-six of the sampled locations are in lavas, dykes and plugs of the old complex. In addition, five sites in the Miocene Flamengos Formation, numbers 47–51, were collected for reference purposes. The great majority of the samples were drill cores oriented by both magnetic and sun compass methods. The site distribution is given in Fig. 1.

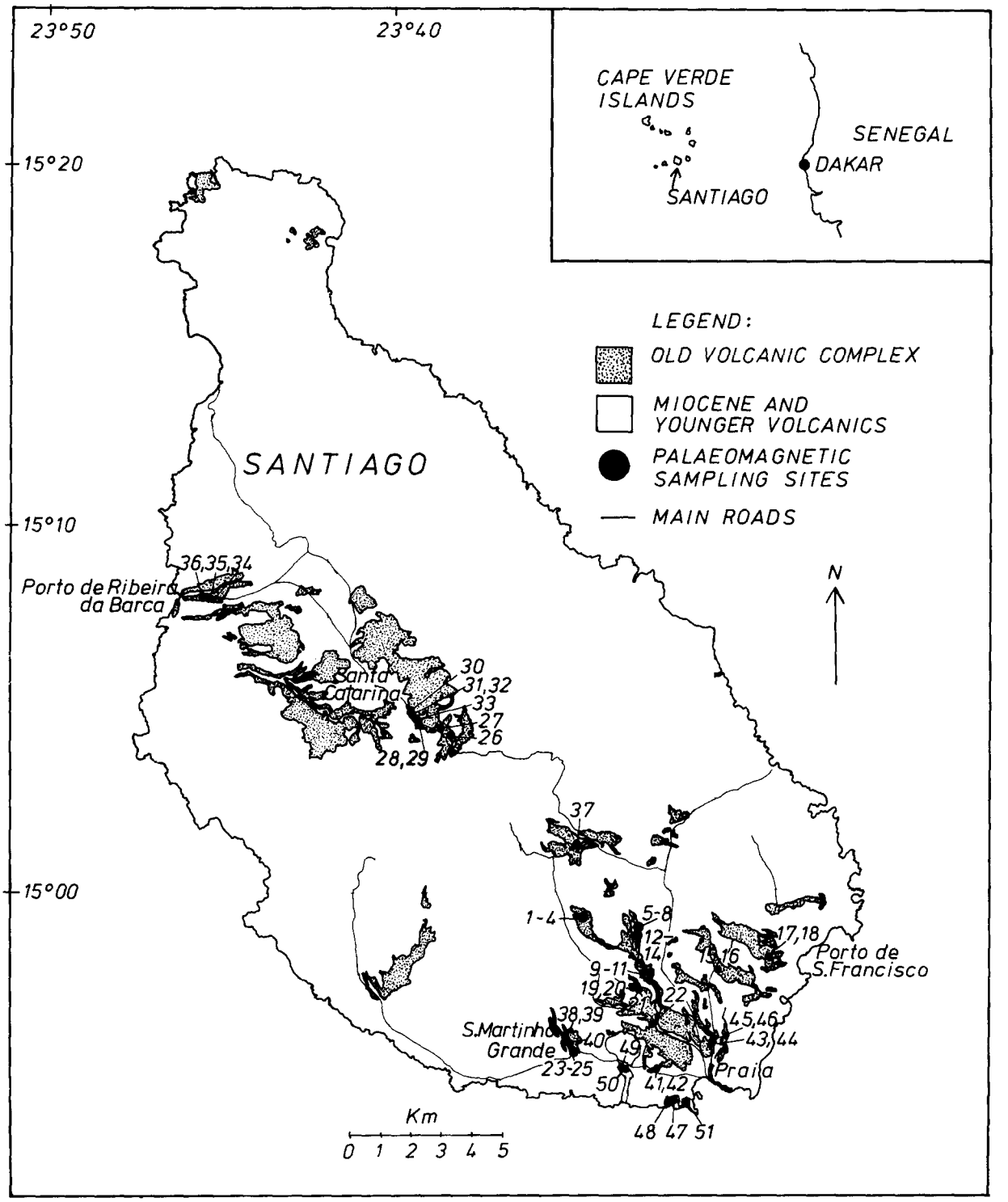


Fig. 1. Geological sketch map of Santiago displaying the surface distribution of the older volcanic complex. Numbers refer to palaeomagnetic sampling sites.

3. Laboratory data

3.1. Methods

Cylindrical specimens, with diameter and height of 1.9 cm, were cut for palaeomagnetic measurements. Demagnetization was either by alternating field (a.f.) using a two-axis tumbler, or by the thermal technique employing a Schonstedt TM-1 as well as a laboratory-built apparatus. After each demagnetization step the remanence parameters of each specimen were determined using a Digico or a Molspin magnetometer. The magnetic minerals of selected samples were studied by reflected light microscopy, as well as by thermomagnetic analysis (saturation magnetization versus temperature, J_s–T). The latter measurements were performed in air on an automatically recording transverse balance, applying a field of 0.46 T and a heating/cooling rate of 20 °C min^{-1}.

3.2. Demagnetization results

The investigated rocks present a typical case of multicomponent magnetization. A broad range of demagnetization data, outlining the magnetization structure at specimen and site level, is displayed in Figs. 3–5. In all, the older rocks (sites 1–46) have a much more complex remanence than the younger rocks (sites 47–51). Therefore, the majority of the examples given in Figs. 3–5 represent the older volcanics.

The predominant direction of magnetization is nearly due N–S with an overall inclination of 20°. In addition, the older complex has a second remanence group with flat-lying inclinations and NNW/SSE declinations. As will be shown below, this latter magnetization can, however, be subdivided into two separate populations distinguished by differences in declination.

Within-specimen/site dual-polarity remanences feature prominently in the older volcanics, as well

FORMATION	ROCK FACIES subaerial	ROCK FACIES marine	AGE
	Alluvium, dunes, flood and slope deposits	Beach sands	Holocene
		Older beach deposits at elevations < 80 m	Pleistocene
Monte das Vacas	Cones of pyroclasts and small lava flows		
Assomada	Basaltic flows and pyroclasts		
Pico da Antónia Complex	Thick series of lava flows (basalts, phonolites and trachytes) with interbedded sediments, pyroclasts and dykes	Congl. and fossiliferous calcarenites interleaved with lavas and pyroclasts	Pliocene
	Trachytes and carbonatites		
Órgãos	Mainly lahars (>100m) with two flood basalts	Congl. and fossiliferous limestones and calcarenites	
Flamengos		Flood basalts, pyroclasts and breccia	Upper Miocene
			tect./erosional unconformity
Old volcanic Complex	5) Carbonatites 4) Phonolitic and trachytic intrusions and extrusions 3) Intravolcanic breccia and breccia type dykes 2) Gabbros, pyroxenites, syenites etc. in irregular bodies and volc. pipes 1) Ankaratrites to basanites in the form of lavas, dykes & plugs	Lava flows	Upper Cretaceous-Lower Tertiary (inferred from present study)

Fig. 2. The stratigraphic column of the Santiago volcanic sequence, simplified after Serralheiro (1976).

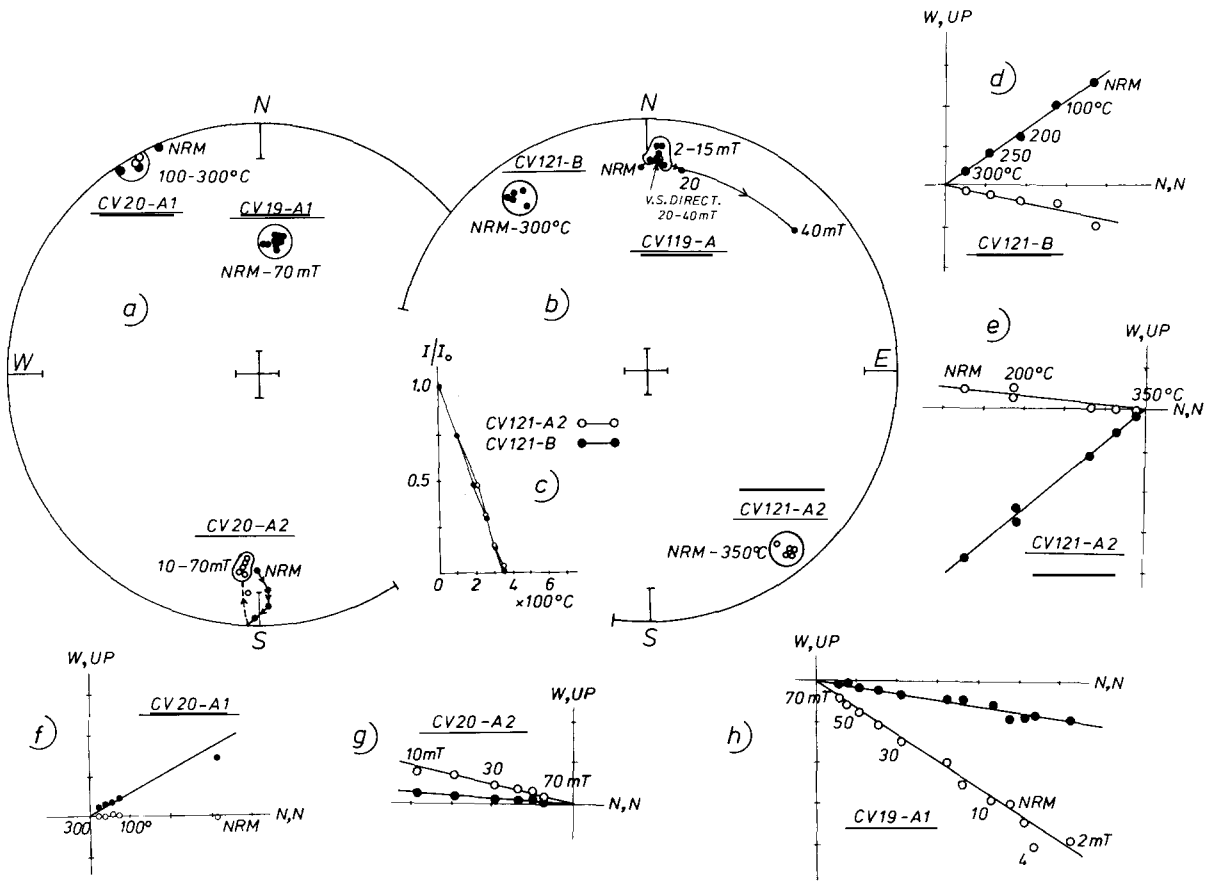


Fig. 3. Examples of specimen demagnetization data from two sampling locations, sites 5 (CV 19-A1, 20-A1 and A2) and 32 (CV 119-A, 121-A2 and B), of the old complex. Note the within-site dual polarity structure, which is a pronounced feature in this complex. Projections (a and b) are equal area, and by convention a lower (upper) hemisphere direction plots as a closed (open) symbol. In the vector diagrams (d–h), projections in the vertical and horizontal planes are marked by open and closed symbols, respectively. The vector subtracted (v.s.) directions, between the 20 and 40 mT demagnetization steps, for specimen CV 119-A, is shown by a triangle. See text for further discussion of results.

as in one site of the younger rocks (site 47, cf. specimens CV 168-A1 and B1, Fig. 5a and d). Figure 3a, f, g and h display the remanence data for three specimens of site 5; two of these samples, CV 19-A1 and 20-A2, accord with the dominant N–S magnetization axis, while the third sample contains the shallow northwesterly remanence. Figure 3b, c, d and e depict a case (site 32) where practically complete antiparallelism of remanence directions is recorded for two adjacent specimens, CV 121-A2 and 121-B, that possess the flat-lying NNW–SSE axis.

Cases of multicomponent palaeomagnetism will normally include many samples that have magnetic properties that prevent reliable component separation. In the present study 28% of the investigated specimens, mostly concentrated in particular sites (numbers 9–11, 18–20, etc.), fall into this category. During progressive demagnetization most of these specimens define segments of remagnetization circles between the characteristic directions established by other samples, for example CV 156-A1 and CV 174-A2, Fig. 5b, c and e. However, the absence of a window in the demagnetization spectra of such specimens, in which only one of the components is being erased, and the frequent occurrence of an erratic behaviour at higher fields/temperatures, prior to reaching an

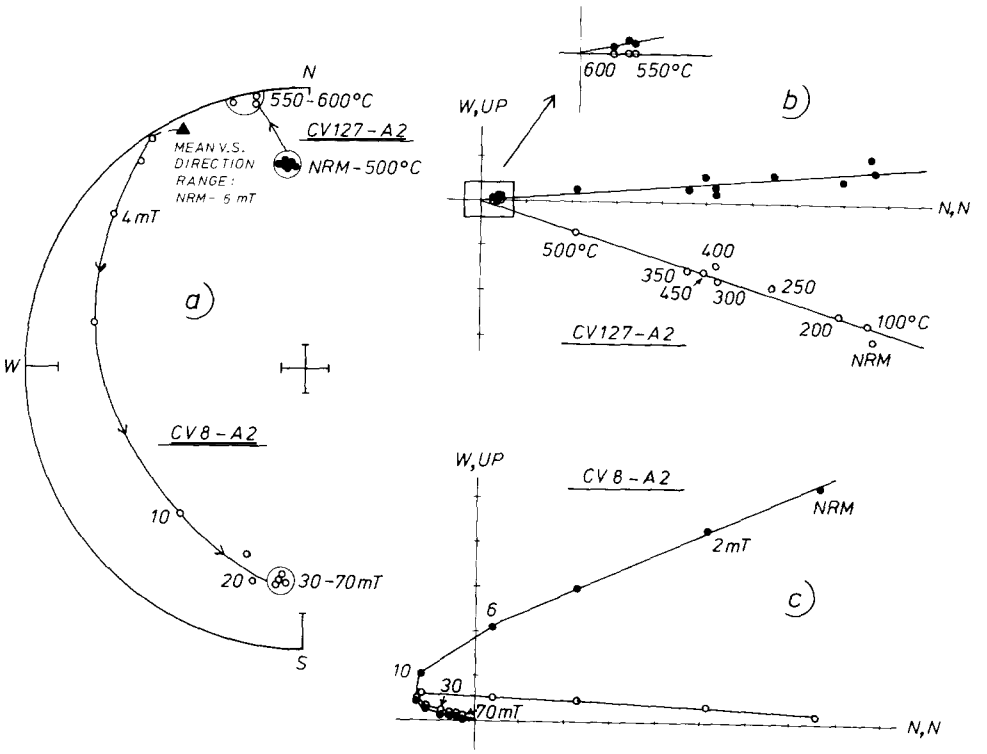


Fig. 4. Further specimen demagnetization results from the old Santiago complex. Conventions as for Fig. 3.

end-point magnetization, preclude the experimental/analytical separation of their inferred single component directions. These samples are consequently not included in the palaeomagnetic database given below, but their general directional behaviour serves to give further confidence to the characteristic results defined by the bulk of the investigated material.

A total of 92 individual remanence directions have been determined from 37 sites in all. All of these palaeomagnetic results constitute linear segments in orthogonal vector diagrams (of which several examples are shown in Figs. 3–5). All directions are plotted in Fig. 6. By inverting the reversed directions it is seen that the results of the older complex, Fig. 6c, tend to define three different groups discriminated by direction, called A1, A2 and B, while the data from the younger lavas, Fig. 5d, are entirely confined to the main group, the A1 population.

3.3. Magnetomineralogy

The observed multiphase magnetization, a structure very different from what should be found if these rocks had essentially retained their original cooling remanence, should somehow be reflected in the texture and chemical composition of the oxide mineralogy. From the thermal demagnetization studies one recognizes that the remanence intensity frequently decays to zero at around 350 ° C. Such observations suggest that maghemite is an important remanence carrier in these rocks.

The opaque minerals of 10 investigated polished sections (scattered throughout the collection) are dominated by euhedral titanomagnetite (TM) grains in high-temperature oxidation class I (Ade-Hall et al., 1971). Free ilmenite has not been observed. Some of the larger TM particles contain occasional ilmenite lamellae. However, the absence of a trellis pattern of these lamellae suggests

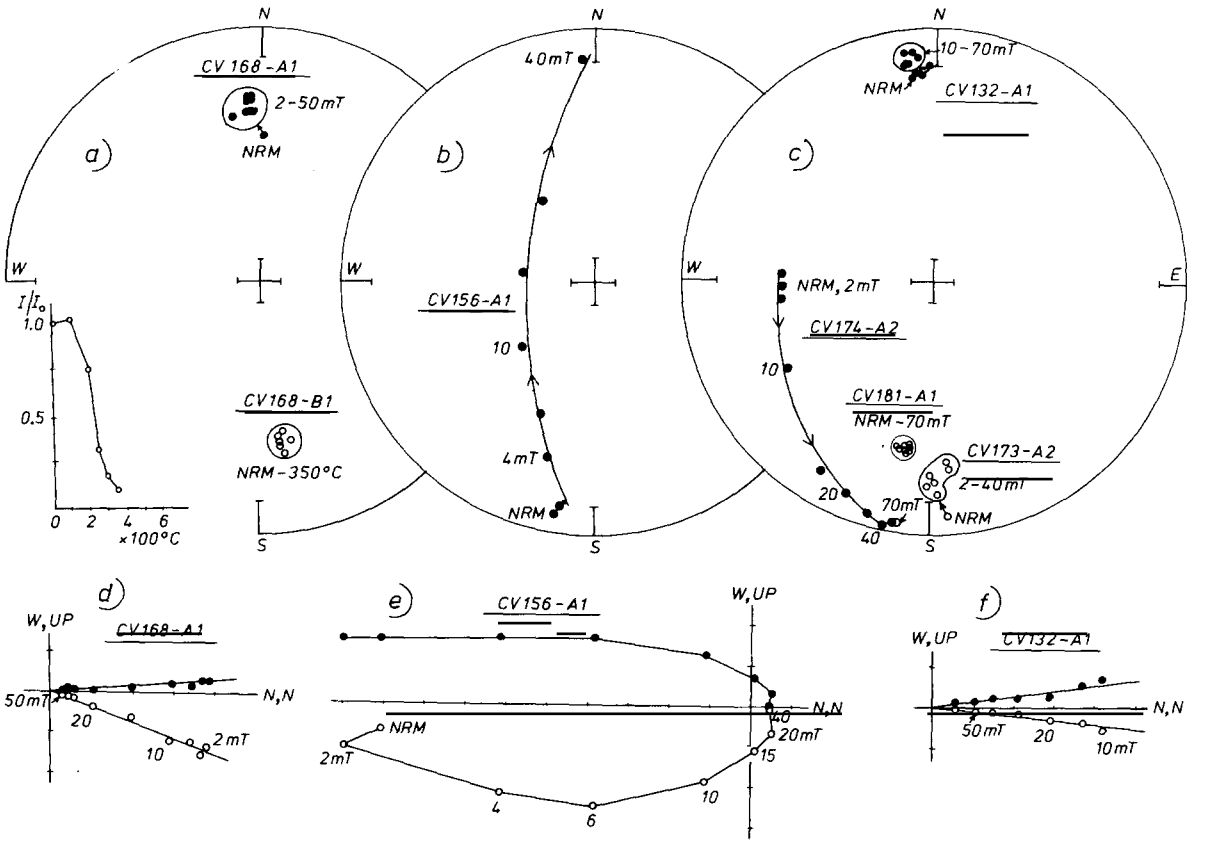


Fig. 5. Further demagnetization results from the Santiago volcanics. Specimens from drill cores CV 168, 173, 174 and 181 are from the base of the younger volcanic sequence (Upper Miocene), while the remaining samples represent the older volcanics. Note the component overlap of the normal and reverse magnetizations for specimen CV 156-A1 (b and e) revealed by the continuously changing directions of erased vectors versus progressive demagnetization. Many specimens from the older volcanics show this latter type of behaviour. Diagram convention as for Fig. 3.

a sandwich-growth rather than an exsolution origin due to deuteric high temperature oxidation. The colour of the TM grains range from brown-tan to grey. The lighter coloured grains, observed in most of the samples, are indicative of varying degrees of low-temperature oxidation, a suggestion that is further supported by the presence of contraction cracks (Akimoto et al., 1984) and granulation (Ade-Hall et al., 1971). Figure 7 displays characteristic examples of these low-temperature oxidation features. The groundmass varies from apparently fresh (in the younger rocks) to strongly oxidized (red-brown in colour).

All thermomagnetic curves show marked irreversible characteristics as shown in Fig. 8. Sample CV 174-A2 (from a young lava, site 49) has a low initial Curie temperature (T_c) around 180 °C, indicative of almost unaltered titanomagnetite. This is in accordance with the microscopic evidence that shows euhedral, tan coloured TM grains without visible oxidation features. Upon heating to 700 °C, the TM grains apparently undergo an oxy-exsolution process producing nearly pure magnetite. Apart from this example of 'virgin' state mineralogy, all other examples of Fig. 8 tend to express the inversion of structurally metastable titanomaghemites at different stages of oxidation (O'Reilly, 1983). Thus, sample CV 150-A1 (site 42) has an initial T_c of ~ 125 °C, but an unstable mineralogical state is already seen in thermal treatment

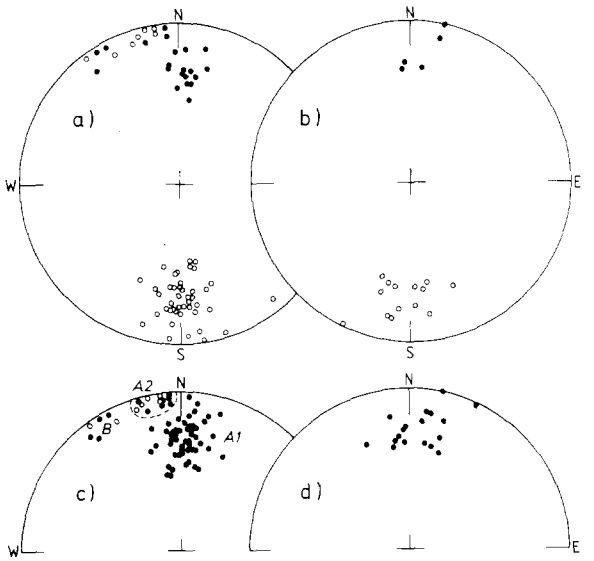


Fig. 6. Distribution of characteristic remanence directions recovered from samples of the older (a) and the younger (b) volcanic succession of Santiago. Diagrams (c) and (d) display the equivalent data but with all reversed directions inverted. The characteristic magnetization of the younger volcanic succession correspond to the inferred youngest palaeomagnetic component (A1) of the older complex. Diagram conventions as for Fig. 3.

around 400 °C, causing irreversible thermomagnetic behaviour on cooling from 410 °C. The overwhelming demonstration of a strong magnetic phase, probably nearly pure magnetite, produced after heating to 700 °C, suggests the original presence of slightly oxidized titanomaghemite, forming a titanium-poor TM as the important magnetic inversion product. The heating curve of CV 142-A1 (site 39) defines initial T_c values at around 300 °C and 510 °C, respectively. Again, we believe that the total J_s–T behaviour of this sample reflects the presence of maghemite, having been subjected to a stronger low-temperature oxidation in nature than sample CV 150-A1. Samples CV 178-C2 (site 51) and CV 129-A1 (site 35) display reduced high-field magnetization after cooling, indicative of a substantial production of hematite at high temperatures. The two thermomagnetically irreversible inflections (below 500 °C) for sample CV 178-C2 suggest continuous breakdown of a strongly magnetic phase, probably a highly oxidized titanomaghemite. The Curie temperature around 580 °C may therefore reflect the inversion product during heating rather than being an inher-

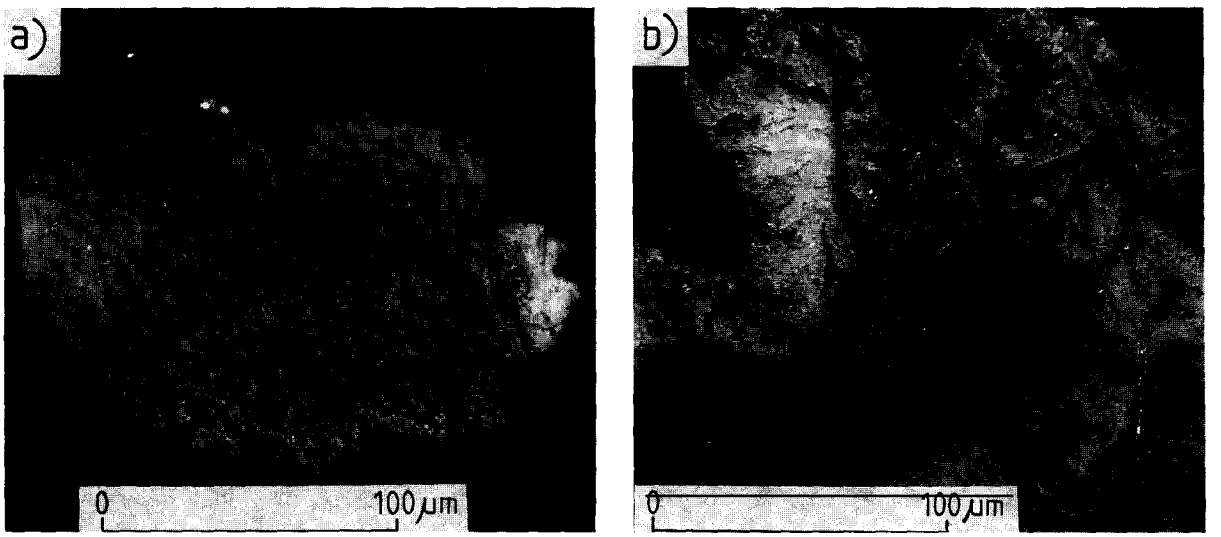


Fig. 7. Photomicrographs of opaque grains in polished sections of the older volcanic complex depicting characteristic low-temperature oxidation features of titanomagnetite grains: (a) advanced stage of granulation (CV 38, site 10); (b) typical patterns of contraction cracks caused by maghemitization (CV 123, site 33).

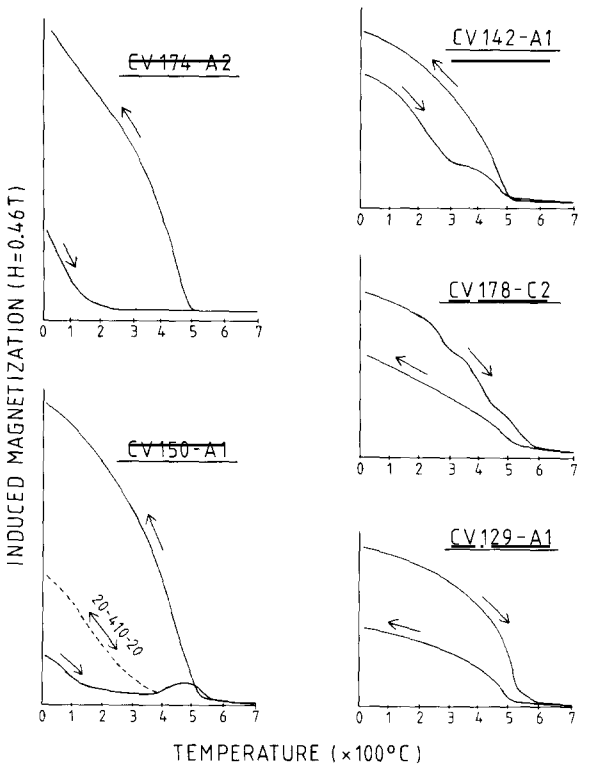


Fig. 8. Characteristic curves of saturation magnetization versus temperature (J_s–T). Suggested interpretation of these data, aided by microscopic observations, is as follows: CV 174-A2 (site 49), unoxidized titanomagnetite; CV 150-A1 (site 42), CV 142-A1 (site 39), CV 178-C2 (site 51) and CV 129-A1 (site 35) have all suffered varying degrees of maghemitization, and the samples are listed in order of inferred increasing oxidation. Consult also Fig. 7 for optical mineral information of CV 150-A1 and CV 142-A1.

ent property of the unheated sample. We infer that sample CV 129-A1 has already undergone this inversion process in nature.

In conclusion, the combined microscopic and thermomagnetic evidence suggest that sample CV 174-A2 (site 49) has retained its original magneto-mineralogy, while all other samples of Fig. 7 have been subjected to a varying degree of low-temperature oxidation producing cation deficient Fe–Ti oxides. With increasing state of oxidation the mineralogically altered samples of Fig. 8 can be arranged in the following sequence: CV 150-A1 (least altered), CV 142-A1, CV 178-C2, and CV 129-A1 (most altered). The field appearance, the palaeomagnetic structure, microscopy, and thermomagnetic evidence all suggest that the Flamengos lavas (sites 47–51) are the least altered rocks of the collection. These younger rocks still carry original titanomagnetite as the predominant magnetic mineral, maghemite having only a patchy occurrence. On the contrary, the older complex has been subjected to a penetrative low-temperature alteration, and cation deficient Fe–Ti oxides, in different stages of oxidation, now represent the prominent magnetic constituents.

4. History of remanence acquisition; age perspectives

The laboratory data outlined above suggest that the Flamengos Formation of Upper Miocene age has to a large extent retained its original magnetic mineralogy. Maghematization and a dual-polarity remanence have been clearly recognized in only one of the five sites investigated, so it is assumed that the Flamengos lavas carry essentially their original cooling magnetization. On the contrary, the older volcanic complex has suffered a penetrative low-temperature alteration leading to a general maghemitization of the original iron–titanium oxides. The fossil magnetization of the older rocks has indeed a fairly complex structure; three palaeomagnetic axes can be distinguished, and each of them is defined by both normal and reverse components. This suggests that the older complex has had a varied magnetization history. Figure 6c and d display the palaeomagnetic composition of the older complex and the Flamengos lavas, respectively. The overall palaeomagnetic data, with respect to individual axes, are listed in Table 1.

The three magnetization groups displayed by the old complex (Fig. 6c) do not overlap, and the various populations are significantly different at the 95% probability level. In view of the conspicuous post-Upper Miocene volcanism in Santiago, providing an effective thermo-chemical basis for magnetic resetting of the older rocks, it is not surprising that the predominant magnetization of the older complex, the A1 group, is indistinguishable from the characteristic magnetization of the Flamengos lavas. On the whole, the penetrative

TABLE 1

Summary of palaeomagnetic results [a]

Formation	Palaeomagnetic group	N	D_m	I_m	α_{95}	K	Pole
Flamengos	A1	19	004	16	5.4	18.5	80°S, 312°E
Older Complex	A1	55	002	20	2.6	52.9	83°S, 320°E
Older Complex	A2	12	351	01	2.7	230.7	71°S, 006°E
Older Complex	B	6	328	−01	5.0	134.8	54°S, 041°E

[a] N = number of specimen vectors, D_m = mean declination, I_m = mean inclination, α_{95} = half-angle of the cone of 95% confidence about the mean (in degrees) and K = precision parameter (Fisher, 1953).

low-temperature oxidation of the old complex has not developed to an extent that would significantly enhance magnetic stability through reduction in effective grain size. The alteration of the great majority of original TM grains has not gone beyond the stage of maghemitization, causing an irregular pattern of conchoidal contraction cracks (Fig. 7). However, the grains of these subdivided assemblages still constitute large multidomain particles, susceptible to thermoviscous magnetization and further oxidation processes. Granulation, which apparently is a more effective mechanism for production of a resistant magnetomineralogy, is only moderately developed in these rocks. Therefore, the inadequate magnetomineralogical state of the older volcanics, versus the considerable post-Upper Miocene volcanic activity (with strong remagnetization potential), does not provide a very promising basis for detecting the original magnetization(s). However, as argued below, two of the magnetizations uncovered, the A2 and B components (Fig. 6c), are consistent with pre-Upper Tertiary ages.

It is interesting to note that the two main directional groups, A1 and A2, can be distinguished in the data of the earlier palaeomagnetic investigation (Watkins et al., 1968), plotted in Fig. 9. In the present study the specimen population of individual sites shows the whole range of palaeomagnetic variability. Also, if we had followed a procedure of straightforward computation and analysis of site mean directions, as applied by Watkins et al., we would have failed to recognize the B magnetization, and even our A2 remanence would have been blurred. This is owing to the fact that the A1 magnetization is the predominant remanence component in the majority of sites. Therefore, the only explanation we can suggest for the indication of a dual magnetization structure, i.e. the A1 and A2 magnetizations, in the site mean data of Watkins et al., (these authors did not, however, distinguish between two palaeomagnetic populations), is that by coincidence they have collected a number of sites with a preponderance of the A2 component.

Figure 10 depicts the three Santiago palaeopoles (A1, A2 and B) in conjunction with palaeomagnetic poles from the island of Maio (Cape Verde Islands) and the Canary Islands, and the Meso-Cenozoic apparent polar wander (APW)

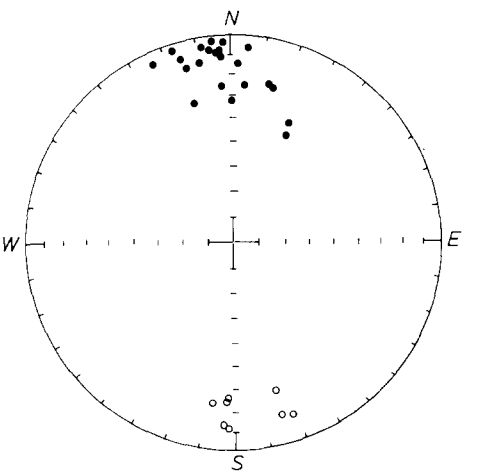


Fig. 9. Site mean direction of magnetization of Santiago volcanics obtained by Watkins et al. (1968). According to their sampling locations most of the sites represent the old volcanic complex. Note that the magnetization directions tend to form two remanence populations, i.e. the A1 and A2 group as defined by the present study (Fig. 6c). Diagram conventions as for Fig. 3.

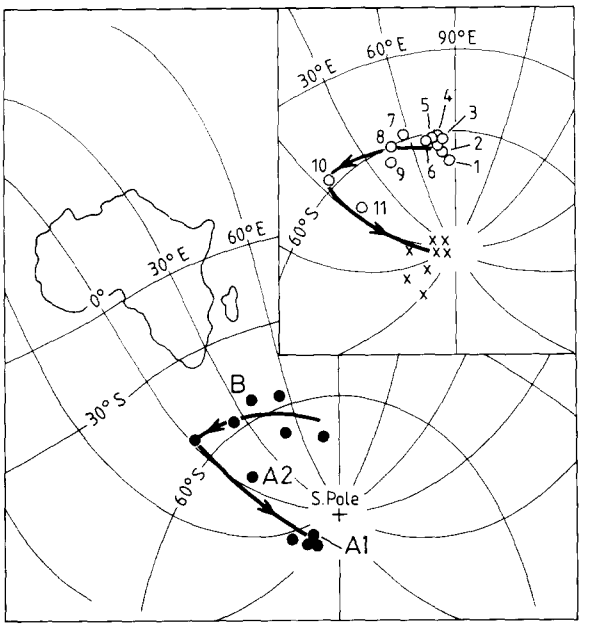


Fig. 10. Santiago palaeomagnetic poles (present study) in association with poles (closed circles) for Maio (Cape Verde Islands) (Storetvedt and Løvlie, 1983) and the Canary Islands (Storetvedt et al., 1979; Soretvedt, 1980). The Cape Verde and Canary Islands data are compared with the Meso-Cenozoic reference polar path for continental Africa (inset diagram). Crosses represent late Tertiary data, and the numbered poles are as follows: 1, Mlanje (Gough and Opdyke, 1963; Briden, 1967); 2, mean Mesozoic Southwest Africa (Hailwood and Mitchell, 1971); 3, Lupata, 106 Ma (Gough and Opdyke, 1963; Gough et al., 1964); 4, Shawa ijolite (Gough and Brock, 1964); 5, mean Mesozoic Northwest Africa (Hailwood and Mitchell, 1971); 6, Hoachanas (Gidskehaug et al., 1975); 7, Kimberlite pipe, ~86 Ma (McFadden and Jones, 1977); 8, Kimberlite pipes, 90 Ma (Hargraves and Onstott, 1980); 9, Wadi Natash volcanics, 81–90 Ma (El Shazly and Krs, 1973); 10 and 11, Lower-(Middle) Tertiary volcanics, Egypt (Gouda Hussain et al., 1979).

path for continental Africa. The APW branch that progresses in a westerly direction close to the 60°S parallel was previously considered to be of Upper Cretaceous age (Storetvedt, 1980; Storetvedt and Løvlie, 1983), but recent work (Storetvedt et al., 1990) suggests that the part of the APW path that falls to the west of ~60°E dates from the Lower Tertiary. Combined palaeomagnetic and climatic evidence suggests further that the final polar pattern, i.e. the curve from the cusp at ~0°E, 55°S (Fig. 10) to near the present pole, formed rapidly around the Eocene–Oligocene boundary (Storetvedt, 1990; Storetvedt et al., 1990). These 'calibrations' of the African polar wander path lead us to conclude that the two oldest Santiago magnetizations recorded, A2 and B, are both of Lower Tertiary age.

5. Conclusions

From the present study of Santiago volcanic rocks the following conclusions can be drawn.

(1) Reflection microscopy of rocks from the old complex shows an apparent absence of ilmenite lamellae in the titanomagnetite (TM) grains, suggesting that high-temperature oxidation is at most only developed to a very modest extent. Therefore, the original TM grains were fairly vulnerable to low-temperature alteration, and the magneto-mineralogical evidence, as well as the remanence properties, concur with this assumption. The present characteristic remanences, notably the B and A2 components, are apparently carried by titanomaghemites, and a lengthy magnetization history can be envisaged.

(2) The oldest magnetization recorded, the B component, seems to have been acquired in Palaeocene–Eocene time, while the A2 remanence is likely to date from around the Eocene–Oligocene boundary. Both the B and the A2 magnetizations must be regarded as of secondary origin, having been impressed through low-temperature oxidation sometime after the original cooling (see also below). The two Lower Tertiary magnetizations became strongly overprinted by the A1 remanence, probably associated with the pronounced Neogene volcanism of the island. Prior to the latter overprinting, the B and A2 components were probably much more apparent than they are today, and remains of the original thermoremanent magnetization (TRM) might have been present as well. However, judging from the strong Neogene volcanism, its associated hydrothermal activity is likely to have provided the conditions for extensive thermoviscous and chemical remagnetizations, so that any original TRM was probably erased altogether. Only a minor fraction of the Lower Tertiary secondary magnetizations, the

A2 and B components, have escaped the Neogene magnetic resetting.

(3) Faced with the evidence of a secondary Lower Tertiary origin for the oldest Santiago magnetization, the actual age of the older complex may well date from the Upper Cretaceous. It seems reasonable to link the geological history of Santiago with that of the nearby island of Maio (Serralheiro, 1970; Stillman et al., 1982; Storetvedt and Løvlie, 1983) and the island of Fuerteventura in the Canary Archipelago describing an evolutionary scheme that seems to mirror the development of the entire northwest African continental margin (Storetvedt, 1980, 1987). This model implies a tectonomagmatic uplift of deep-sea crust, that locally arose to subaerial level, at around Turonian (~ 90 Ma ago), forming the intrusive cores of individual islands. A major time span of volcanic quiescence and erosion elapsed (during which the Santiago B and A2 magnetizations evolved) before extensive volcanic activity resumed in the Upper Neogene.

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